

A Reproducible Control-Validation Workflow for Circular-Shift and Surrogate Tests of Galactic Radial Profiles

Vitor M. F. Figueiredo^{a,*}

^a*Independent Researcher, Nisa, Portugal*

Abstract

Correlations between radially organized covariates and the residuals of galaxy rotation-curve fits are used as diagnostics of inner-region structure. Assessing such correlations requires controls that account for the strong radial organization of the underlying data, because a correlation can exceed a simple label-shuffle comparison while reflecting only a broad, smooth radial trend. We present a reproducible, branch-separated control-validation workflow for radial-structure diagnostics in SPARC-like rotation-curve data. For a partial-rank branch of 134 galaxies, a radius-conditioned partial-rank statistic is compared against stochastic shuffle/permutation empirical control distributions and against exact circular-shift randomization distributions that preserve radial ordering. The observed absolute partial-rank statistic has across-galaxy median $\simeq 0.509$; the shuffle control median-of-medians is $\simeq 0.182$, whereas the ordering-preserving exact circular-shift control median-of-medians is $\simeq 0.532$, comparable to the observed statistic. For a separate spectral-control branch of 76 galaxies, an amplitude-based spectral-concentration statistic is evaluated against amplitude-preserving random-phase controls; because the statistic is amplitude-based, equality with these controls holds by construction and is reported as a caveat rather than as evidence. Fixed smooth profiles are included as deterministic comparators, not stochastic controls. We document the workflow, its deterministic-seed implementation, and an audited descriptive demonstration for reuse, and make no physical interpretation or model-validation claim.

*Corresponding author.

URL: <https://orcid.org/0009-0004-7358-4622> (Vitor M. F. Figueiredo)

Keywords: methods: statistical, methods: data analysis, galaxies: kinematics and dynamics, surrogate data, permutation testing, reproducibility

1. Introduction

Radial-structure diagnostics correlate a radially organized covariate with the residuals of galaxy rotation-curve fits in order to probe inner-region structure. Rotation curves and radial profiles are widely used observational diagnostics in galaxy studies, including broad rotation-curve reviews, high-resolution nearby-galaxy rotation-curve surveys, and galaxy-mass reviews [11, 2, 1]. A practical difficulty is that the radius itself, and any smooth radially organized covariate, can induce apparent correlations that do not reflect localized structure. Establishing whether a measured correlation carries information beyond broad radial organization is therefore primarily a computational and statistical problem of control design.

Permutation and surrogate procedures are established tools in general statistical and correlation-testing contexts, with related resampling and non-parametric methods widely discussed in astronomy-statistics practice [10, 5, 3, 7, 14]. They also have a lineage in surrogate-data testing of an observed statistic against randomized alternatives [12]. These procedures are general-purpose; we do not claim that any particular control family is common published practice in rotation-curve analysis. Different control families preserve different structure, and a correlation that exceeds a label-shuffle comparison may still be reproduced by controls that preserve radial ordering.

In this work we present a reproducible, branch-separated control-validation workflow for radial-structure diagnostics in SPARC-like rotation-curve data [6]. The implemented workflow is referred to here as the Nisa Radial Control Battery (NRCB). The workflow evaluates a radius-conditioned partial-rank statistic against stochastic shuffle/permutation empirical control distributions and against exact circular-shift randomization distributions, and evaluates an amplitude-based spectral-concentration statistic against amplitude-preserving random-phase controls and fixed smooth references. Our contribution is the design of this control battery, its deterministic-seed implementation, and an audited descriptive demonstration on two explicitly separated galaxy samples. We make no physical interpretation or model-validation claim; the demonstration characterizes how the control families behave, not the astrophysics of any galaxy.

The paper is organized as follows. Section 2 defines the data context and the two sample branches. Section 3 defines the diagnostic statistics and control families. Section 4 presents the descriptive demonstration. Section 5 states limitations, Section 6 documents reproducibility, and Section 7 concludes.

2. Data and branch definition

SPARC provides the rotation-curve data context and comprises 175 galaxies [6]. The computational workflow uses two explicitly separated samples: an $N = 134$ partial-rank branch and an $N = 76$ spectral-control branch. The branch sizes are not interchangeable.

2.1. Branch membership

The frozen partial-rank branch required completion of the reconstruction contract with at least eight valid radii, finite required vectors, positive radial and velocity fields, a finite radial-template covariate, and a computable radius-conditioned statistic. The statistic-level implementation separately requires at least five finite joint points.

The spectral-control branch is the subset with at least 16 reconstructed points, at least 16 finite unique positive-radius samples, finite interpolation to a fixed 64-point log-radius grid, and nonzero finite spectral power.

The radius-conditioning coordinate is the dimensionless observed-radius fraction

$$r_{gi} = \frac{R_{gi}}{R_{\max,g}}, \quad R_{\max,g} = \max_i R_{gi}, \quad (1)$$

where R_{gi} is taken from the observed rotation-curve radius column after nonfinite or nonpositive radius and velocity rows have been excluded, and $R_{\max,g}$ is the maximum retained observed radius for galaxy g . The same coordinate is used in the post-reconstruction validity mask and partial-rank residualization. Correctly constructed values lie in $0 < r_{gi} \leq 1$. The adapter also applies a permissive upper check $r_{gi} \leq 1.1$; this check is a post-construction implementation guard and does not redefine the normalization or admit any additional correctly constructed observed-radius fraction.

3. Methods

3.1. Diagnostic definitions

For galaxy g , let $i = 1, \dots, n_g$ index its frozen valid radial bins. Let r_{gi} denote the radius-conditioning coordinate, $y_{gi} = |\log_{10}(v_{\text{rec},gi}/v_{\text{obs},gi})|$ the residual magnitude, and σ_{gi} the reconstructed total radial covariate.

3.1.1. Radial-template covariate and partial-rank statistic

The normalized covariate and radial-template function are

$$c_{gi} = \frac{\sigma_{gi}}{\text{median}\{\sigma_{gj} : \sigma_{gj} > 0, \sigma_{gj} \text{ finite}\}}, \quad (2)$$

and

$$t(c) = \text{clip}_{[0,1]} \left[\exp \left\{ - \left(\frac{\log_{10}(\max(c, 10^{-12}))}{0.50} \right)^2 \right\} \right]. \quad (3)$$

A scale fallback of one is used when no finite positive covariate value exists, and nonfinite normalized entries are replaced by zero.

Let $\text{rank}(\cdot)$ assign average ranks to ties and $X_g = [\mathbf{1}, \text{rank}(r_g)]$. For any vector z_g , define

$$\hat{\beta}_z = \arg \min_{\beta} \|\text{rank}(z_g) - X_g \beta\|_2^2, \quad u_z = \text{rank}(z_g) - X_g \hat{\beta}_z, \quad (4)$$

implemented through least-squares fitting. The signed statistic is

$$\rho_g = \text{corr}(u_{t(c)}, u_y), \quad (5)$$

and the production-facing headline summary uses $|\rho_g|$.

3.1.2. Spectral concentration

Each valid profile is linearly interpolated onto 64 evenly spaced points in $\log_{10}(R/\text{median}_+(R))$ and mean-centered. After a real-valued Fourier transform and removal of the DC component, define

$$p_k = \frac{|F_k|^2}{\sum_j |F_j|^2}, \quad C = 1 - \frac{-\sum_k p_k \log p_k}{\log K}, \quad (6)$$

where K is the number of retained non-DC power bins.

3.2. Control families and empirical summaries

Permutation and surrogate procedures are established tools in general statistical and correlation-testing contexts [10, 7, 14].

3.2.1. Shuffle/permutation controls

For each partial-branch galaxy, $M = 1000$ deterministic-seed realizations permute only the normalized radial covariate. Radius and residual vectors remain frozen, and the template function is applied after permutation. This general baseline does not imply common practice in rotation-curve analysis.

3.2.2. Exact circular-shift randomization distributions

For a galaxy with n_g valid bins, the production-facing ordering-preserving control uses every offset in

$$\mathcal{S}_g = \{1, 2, \dots, n_g - 1\} \quad (7)$$

exactly once. Offset zero is excluded. For each $s \in \mathcal{S}_g$, $c_g^{(s)} = \text{roll}(c_g, s)$ is transformed and evaluated with the same radius-conditioned statistic.

Cyclic wrap-around is an operational finite-sample randomization convention, not a claim that the innermost and outermost radial bins are physically adjacent. Only the normalized radial covariate is rolled; the radius vector, residual profile, finite sample size, and frozen validity mask remain fixed. Applying every non-identity offset preserves the ordered covariate sequence under translation and defines an exact finite randomization space over all nontrivial cyclic alignments. The comparison tests sensitivity to radial ordering and alignment between the radial-template covariate and residual profile, conditional on this operational convention.

3.2.3. Amplitude-preserving random-phase controls

For each spectral-valid residual and radial-template profile, $M = 1000$ controls preserve Fourier amplitudes while randomizing the unconstrained phases at positive frequencies. The mean-centered DC component is zero; the even-length Nyquist coefficient remains real with a randomized sign; and an inverse real-valued Fourier transform restores the profile [12, 10]. More constrained surrogate families can preserve additional data properties [9], but the present random-phase tier is intentionally limited to amplitude preservation and is interpreted only within that control-family definition.

The random-phase comparison is included as a sensitivity diagnostic for the spectral statistic: because spectral concentration is amplitude-based,

amplitude-preserving phase randomization demonstrates which quantities are invariant by construction rather than providing independent physical evidence.

3.2.4. Smooth fixed references

On a 64-point coordinate $x \in [0, 1]$, the fixed profiles are

$$\exp(-x), \quad \max(1 - x, 0), \quad \max(1 - x, 0)^2. \quad (8)$$

They are fixed references, not stochastic controls, and receive no empirical percentile or p-value.

3.2.5. Empirical summaries

For finite observed value z_{obs} , finite controls $\{z_m\}_{m=1}^{M_v}$, and $\epsilon = 10^{-12}$, define

$$L = \#\{z_m < z_{\text{obs}} - \epsilon\}, \quad E = \#\{|z_m - z_{\text{obs}}| \leq \epsilon\}, \quad G = \#\{z_m > z_{\text{obs}} + \epsilon\}, \quad (9)$$

with $L + E + G = M_v$. The bounded mid-rank percentile is

$$q = \text{clip}_{[0,1]} \left(\frac{L + E/2}{M_v} \right). \quad (10)$$

For signed or absolute random-tier statistics,

$$p_{\text{MC}} = \text{clip}_{[0,1]} \left[\frac{1 + \#\{|z_m| \geq |z_{\text{obs}}| - \epsilon\}}{M_v + 1} \right]. \quad (11)$$

The same formula is used for the exhaustive nontrivial circular-shift set. For the nonnegative phase-blind spectral statistic, $p_{\text{spectral}} = \min\{1, 2 \min(q, 1 - q)\}$. Fixed references receive neither an empirical percentile nor an empirical p-value.

4. Demonstration

Figure 1 presents the branch-separated computational workflow. Shuffle/permutation controls define stochastic empirical control distributions. Circular-shift controls instead use exhaustive nontrivial radial offsets to define exact circular-shift randomization distributions.

Control workflow for radial-structure diagnostics

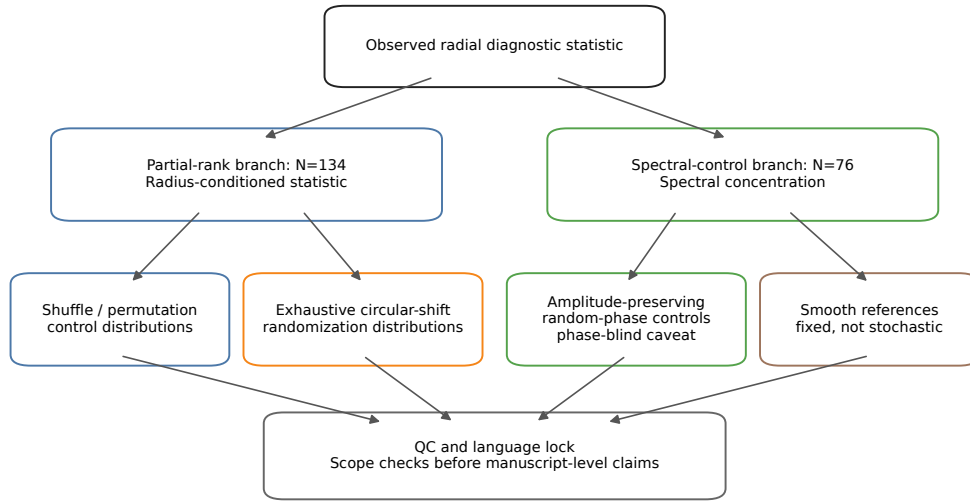


Figure 1: Control workflow for radial-structure diagnostics. The workflow separates the $N = 134$ partial-rank branch from the $N = 76$ spectral-control branch. Shuffle/permutation controls form stochastic empirical control distributions, whereas circular-shift controls use exhaustive nontrivial radial offsets to define exact circular-shift randomization distributions. Amplitude-preserving random-phase controls carry a phase-blind/by-construction caveat. Smooth profiles are fixed references, not stochastic controls.

For the $N = 134$ partial-rank branch, the observed absolute statistic has median $\simeq 0.509$ (IQR 0.280–0.773). Per-galaxy shuffle-control medians have median $\simeq 0.182$ (IQR 0.135–0.232), while exact circular-shift medians have median $\simeq 0.532$ (IQR 0.340–0.649). Across galaxies, the corresponding empirical p-values have medians $\simeq 0.026$ and 0.500, respectively. These summaries are not pooled-across-galaxy p-values.

For the separate $N = 76$ spectral branch, residual and radial-template spectral concentrations have medians $\simeq 0.676$ (IQR 0.589–0.755) and $\simeq 0.754$ (IQR 0.663–0.817), respectively. Random-phase control medians are identical at displayed precision by construction. The exponential, linear, and quadratic fixed-reference values are approximately 0.548, 0.553, and 0.570.

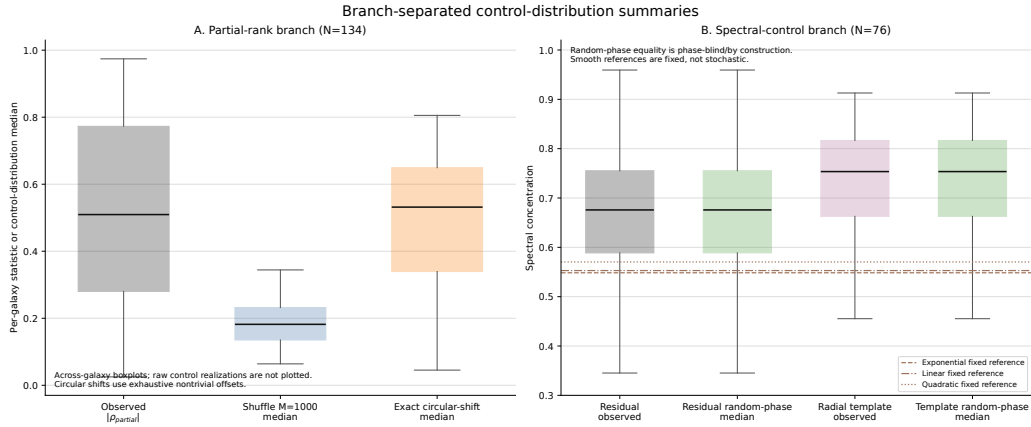


Figure 2: Branch-separated control-distribution summaries. Panel A shows the $N = 134$ partial-rank branch and per-galaxy medians from the $M = 1000$ shuffle empirical control distributions and exhaustive circular-shift randomization distributions. Panel B separately shows the $N = 76$ spectral-control branch, random-phase control medians, and smooth-profile fixed references. Panels summarize each per-galaxy control distribution by its median; empirical percentiles and p-values remain in the production summary table, and the full raw realizations are not plotted. Fixed references are not stochastic controls.

For both spectral targets, per-galaxy mid-rank empirical percentiles are 0.500 and doubled equal-tail quantities are 1.000. This follows from amplitude preservation and the definition of spectral concentration.

5. Discussion and limitations

No close rotation-curve or radial-profile control precedent was verified; the citation scope used here is general-methodological. Cyclic wrap-around is an

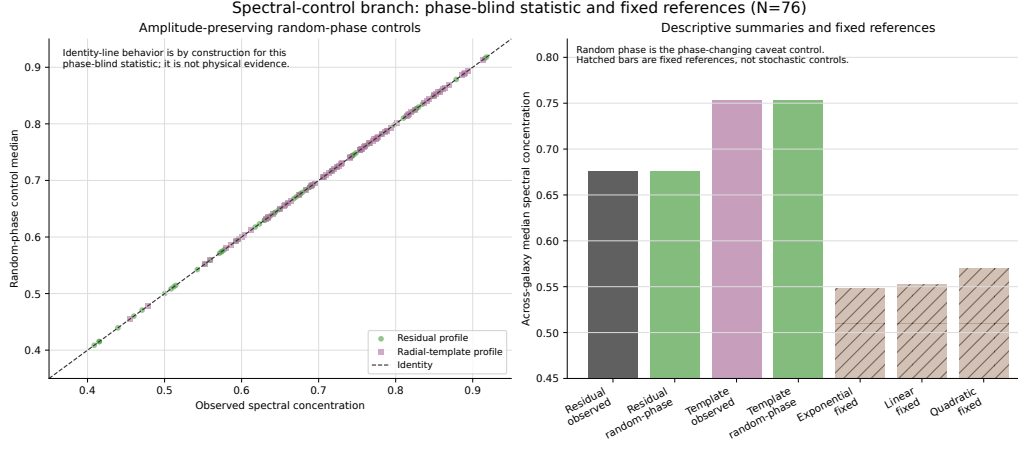


Figure 3: Phase-blind spectral-concentration caveat and fixed references ($N = 76$). Panel A compares observed spectral concentration with per-galaxy medians from amplitude-preserving random-phase controls. Identity-line behavior is by construction for the phase-blind statistic and is not independent evidence about the profiles. Panel B shows descriptive across-galaxy medians; smooth profiles are fixed references, not stochastic controls.

operational randomization convention and does not imply physical adjacency of the radial endpoints. Other defensible designs could include alternative detrending, residualization against smooth radial profiles, block/permutation controls, non-cyclic boundary treatments, or alternative radial templates. These alternatives were not tested here. Random-phase spectral equality is by construction, smooth profiles are deterministic comparators, and the two sample branches must remain separated.

6. Reproducibility

The production workflow uses frozen base seed 20260503 and the exact seed namespace `ac-control-validation-m-surrogate-v1`. Randomized seeds are deterministic unsigned integers derived from the first eight bytes of SHA256 applied to the namespace, base seed, canonical galaxy name, tier name, and one-indexed iteration. Python’s built-in hash is not used. Circular shifts and fixed references do not use random seeds.

The release plan follows general scientific-computing and reproducible-computational-research guidance: record executable entry points, inputs, outputs, environment details, and provenance sufficient for independent audit

[13, 8].

The recorded environment includes Python 3.12.7, NumPy 1.26.4 [4], and code commit c8a6c950bd9e. The production entry point is `python src/run_m_surrogate_extension.py -production`. Invalid realizations were retained and counted in emitted totals while excluded from valid empirical statistics. Detailed audit records are retained in repository notes and supplementary provenance files.

7. Conclusion

We have presented the Nisa Radial Control Battery (NRCB), a reproducible control-validation workflow for radial-structure diagnostics in SPARC-like rotation-curve data. The workflow is branch-separated: a partial-rank branch of 134 galaxies and a spectral-control branch of 76 galaxies are analysed independently and are not interchangeable. By comparing each observed statistic against multiple control families, the workflow makes explicit how those families behave. A label-shuffle comparison and an ordering-preserving exact circular-shift comparison can lead to different conclusions about the same statistic; the exact circular shifts act as operational finite-sample controls rather than physical models. For the amplitude-based spectral-concentration statistic, equality with amplitude-preserving random-phase controls holds by construction and is reported as a caveat, and fixed smooth profiles are deterministic comparators rather than stochastic controls.

We make no physical interpretation, discovery, or model-validation claim; the demonstration is descriptive and characterizes control behavior on the two samples. Natural methodological extensions include alternative detrending, non-cyclic boundary conventions, block or permutation controls, additional radial templates, and application to broader datasets.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used OpenAI ChatGPT and Codex, Anthropic Claude and claude code, Google Gemini and Grok for drafting assistance, language refinement, code-review support, and the preparation of audit checklists. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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